Fields and Waves

Exam 2

Spring 2026

1	**/6**
2	**/8**
3	**/8**
4	**/10**
Total	**/32**

Name Solutions

RIN _____

Notes:
SHOW ALL WORK. BEGIN WITH FORMULAS, THEN SUBSTITUTE VALUES AND UNITS. No credit will be given for numbers that appear without justification. Partial credit will be awarded where possible, so show your work so that we can follow your thought process. Use the backs of pages if there is not enough room on the front. If you do, please indicate on the front of your exam that there is more work on the back of pages.

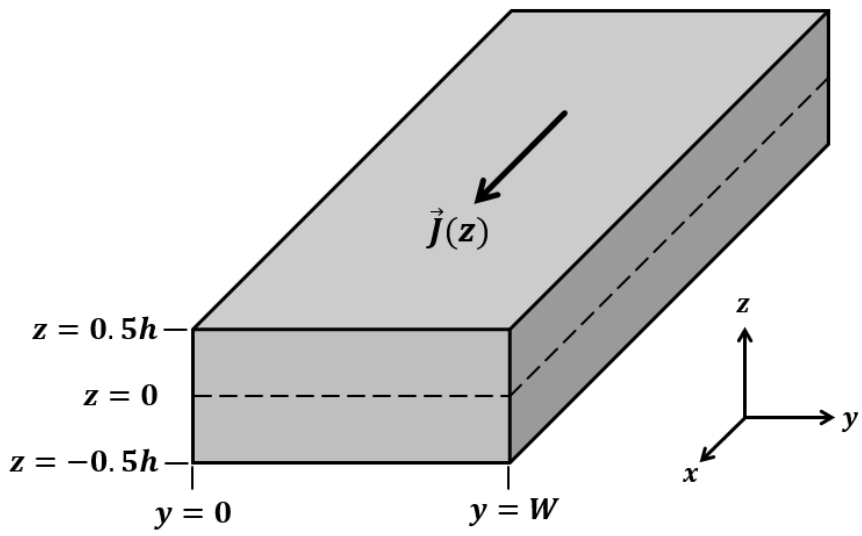
Unless otherwise stated in a problem, provide at least 2 significant digits in answers. Ensure that your calculator is set to use radians for trigonometric functions (and not degrees) and include units with all numerical answers.

At least skim through the entire quiz before you begin and then start with the problems you know best.

Problem 1: Resistance of a Microstrip Line at High Frequencies [6 pts]

As discussed in class, due to the skin effect, current prefers to flow closer to the surfaces of a conductor at higher frequencies. Consider the microstrip line below, which has the following properties:

- Width (in y-dimension): W
- Height (in z-dimension): h
- Length (in x-dimension): infinite
- Conductivity: σ



The current density distribution in the microstrip line is given by:

$$\vec{J}(z) = \begin{cases} J_0\hat{x} & \text{for} \quad 0.4h \le z < 0.5h \\ 0 & \text{for} \quad -0.4h < z < 0.4h \\ J_0\hat{x} & \text{for} \quad -0.5h < z \le -0.4h \end{cases}$$

Ignore any electric field fringing effects due to the finite width of the microstrip line.

a) Calculate the total current I flowing through the microstrip line.

Primary Equation $I = \iint \vec{J} \cdot d\vec{s}$

$$I = \int_{-0.5h}^{-0.4h}\int_{0}^{W}(J_0\hat{x})\cdot(\hat{x}\,dy\,dz) + \int_{0.4h}^{0.5h}\int_{0}^{W}(J_0\hat{x})\cdot(\hat{x}\,dy\,dx) = 0.1h\cdot W\cdot J_0 + 0.1h\cdot W\cdot J_0$$

$$= 0.2hWJ_0$$

$I = 0.2hWJ_0$

b) Calculate the magnitude of the potential drop per unit length V' of the microstrip line.

Primary Equation $V = -\int \vec{E}\cdot d\vec{l}$

$$\vec{E} = \frac{\vec{J}}{\sigma} = \begin{cases} \frac{J_0}{\sigma}\hat{x} & 0.4h \le z < 0.5h \\ 0 & -0.4h < z < 0.4h \\ \frac{J_0}{\sigma}\hat{x} & -0.5h < z \le -0.4h \end{cases}$$

$$V' = V(0) - V(L) = -\frac{1}{L}\int_{L}^{0}\left(\frac{J_0}{\sigma}\hat{x}\right)\cdot(\hat{x}\,dx) = \frac{J_0}{\sigma}$$

$V' = \dfrac{J_0}{\sigma}$

c) Calculate the resistance per unit length R' of the microstrip line.

Primary Equation	$R = \frac{V}{I}$

$$R' = \frac{V'}{I} = \frac{(J_0/\sigma)}{0.2hwJ_0} = \frac{S}{hw\sigma}$$

$$R' = \frac{S}{hw\sigma}$$

d) For the same current density as above, use physical reasoning to explain why increasing the conductivity σ would lead to a decrease in the resistance of the microstrip line.

Since $\vec{J} = \sigma\vec{E}$, increasing σ while $\vec{J}$ is held constant decreases $\vec{E}$. If $\vec{E}$ decreases V' decreases and R decreases.

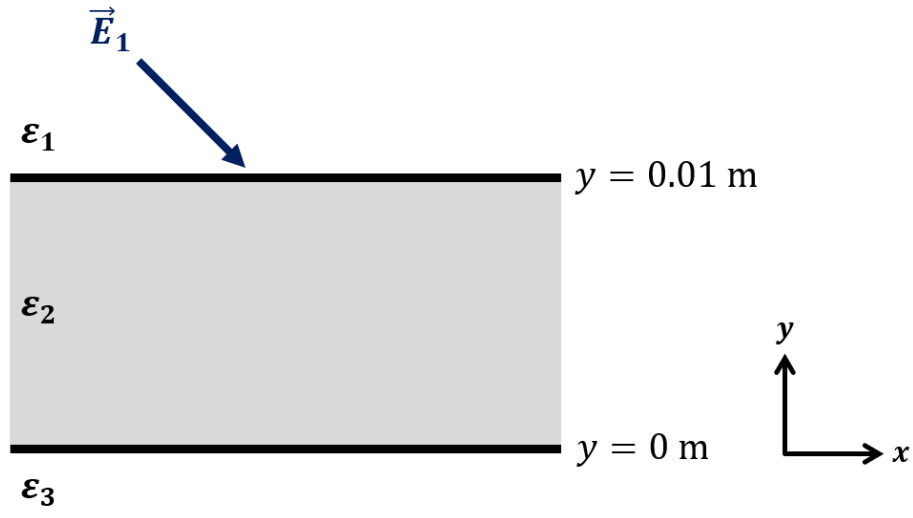
e) For the same total current flowing through the microstrip line, how would the resistance of the microstrip line differ if the current density were distributed uniformly over the cross-section instead of being concentrated near the surfaces? Briefly explain using physical reasoning.

If the total current were held constant, but the current were uniformly distributed, the same total current would flow through a larger cross-sectional area, leading to a lower $\vec{J}$. If $\vec{J}$ decreases, $\vec{E}$ decreases, which leads to a smaller voltage drop and lower resistance.

Problem 2: Electrostatic Boundary Conditions [8 pts]

Shown below is a stack of three dielectric materials:

- **Dielectric 1**: occupies the region $y > 0.01$ m and has permittivity $\varepsilon_1 = 2\varepsilon_0$
- **Dielectric 2**: occupies the region $0 < y < 0.01$ m and has permittivity $\varepsilon_2 = 5\varepsilon_0$
- **Dielectric 3**: occupies the region $y < 0$ and has permittivity $\varepsilon_3 = 2\varepsilon_0$



The electric field in dielectric 1 is given by:

$$\vec{E}_1 = 100\hat{x} - 100\hat{y}$$

a) Find the electric field in dielectric 2, $\vec{E}_2$.

Primary Equations $D_{1n} - D_{2n} = \rho_s$; $E_{1t} = E_{2t}$

material 1 ε_1 / material 2 ε_2 and $\begin{cases} E_{1n} = -100 \text{ V/m} \\ E_{1t} = 100 \text{ V/m} \end{cases}$

- normal fields: $\varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = 0 \rightarrow E_{2n} = \frac{\varepsilon_1}{\varepsilon_2} E_{1n} = \frac{2\varepsilon_0}{5\varepsilon_0}(-100) = -40 \text{ V/m}$
- tangential fields: $E_{1t} = E_{2t} \rightarrow E_{2t} = 100 \text{ V/m}$

$\vec{E}_2 = (100\hat{x} - 40\hat{y})$ V/m

$\vec{E}_2 = (100\hat{x} - 40\hat{y})$ V/m

b) Find the electric field in dielectric 3, $\vec{E}_3$.

$\vec{E}_2 = (100\hat{x} - 40\hat{y})$ V/m $\rightarrow \begin{cases} E_{2n} = -40 \text{ V/m} \\ E_{2t} = 100 \text{ V/m} \end{cases}$

- normal fields: $E_{2n} = \frac{\varepsilon_2}{\varepsilon_3} E_{2n} = \frac{5\varepsilon_0}{2\varepsilon_0}(-40 \text{ V/m}) = -100 \text{ V/m}$
- tangential fields: $E_{3t} = E_{2t} = 100$ V/m

$\vec{E}_3 = (100\hat{x} - 100\hat{y})$ V/m

$\vec{E}_3 =$ $(100\hat{x} - 100\hat{y})$ V/m

c) Calculate the total potential difference across dielectric 2, V_0, (i.e. between $y = 0.01$ m and $y = 0$).

Primary Equation $V = -\int \vec{E} \cdot d\vec{\ell}$

$$V(y=0.01\text{m}) - V(y=0) = -\int_0^{0.01\text{m}} (100\hat{x} - 40\hat{y}) \text{ V/m} \cdot \hat{y}\, dy$$

$$= \int_0^{0.01\text{m}} 40\, dy = 0.01\text{m} \cdot 40 \text{ V/m} = 0.4\text{V}$$

$V_0 =$ 0.4V

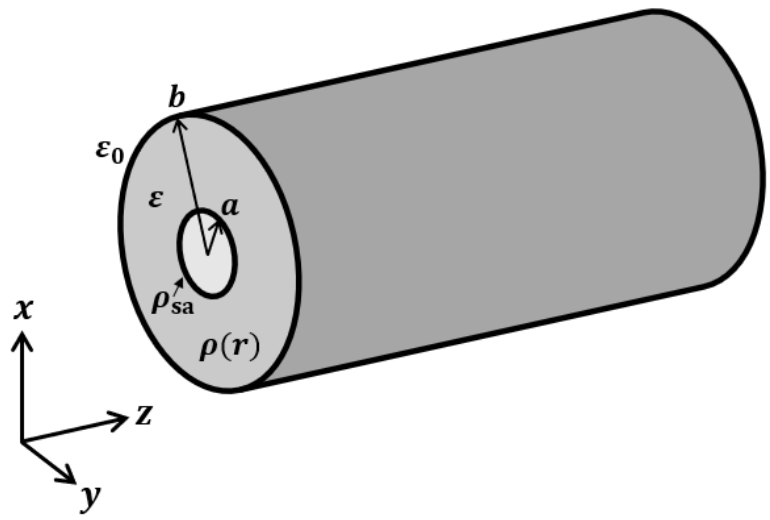
d) Using the concept of polarization in dielectric materials, briefly explain why the magnitude of the electric field in dielectric 2 is different from the magnitudes of the electric fields in dielectric 1 and dielectric 3. Be sure to comment on the relative magnitudes: why is one larger/smaller than the others?

When an electric field is applied to a dielectric material, it causes the molecules to polarize. This generates a response electric field that opposes the direction of the applied field, resulting in a net electric field that is smaller than the applied field. Since $\varepsilon_2 > \varepsilon_1 = \varepsilon_3$, material 2 polarizes more strongly and the total field in dielectric 2 is less than those in dielectrics 1 and 3.

Problem 3: Leaky Coaxial Cable Dielectric [8 pts]

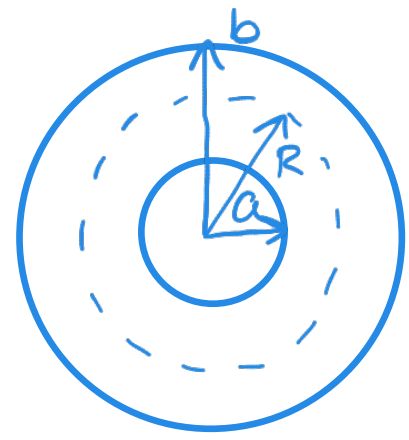
The coaxial cable to the right has a poor dielectric, which has allowed some charge to leak between the inner conductor and outer conductor. The cable has an inner conductor of radius a with surface charge density ρ_{sa} and an outer conducting shell of radius b, held at a potential $V(b) = 0$. The region $a < r < b$ is filled with a dielectric of permittivity ε and volumetric charge density given by:

$$\rho(r) = \rho_0 \frac{a}{r}$$



a) Using Gauss's law, find the electric field $\vec{E}(\vec{r})$ in the region $a < r < b$. Include a diagram that shows the location of your Gaussian surface.

Primary Equation	$\oint \vec{D} \cdot d\vec{s} = Q_{enc}$



$\vec{E}(\vec{r}) = E_r(r)\hat{r}$

flux integral: $\varepsilon \oint \vec{E} \cdot d\vec{s}$

$$\int_0^L \int_0^{2\pi} (E_r(R)\hat{r}) \cdot (R\,d\phi\,dz\,\hat{r}) = 2\pi L \varepsilon R E_r(R)$$

enclosed charge: $Q_{enc} = Q_{surface} + Q_{volume}$

$$Q_{enc} = \int_0^L \int_0^{2\pi} \rho_{sa} \cdot a\,d\phi\,dz + \int_0^L \int_0^{2\pi} \int_a^R \left(\rho_0 \frac{a}{r}\right)(r\,dr\,d\phi\,dz)$$

$$= 2\pi L a \rho_{sa} + 2\pi L a \rho_0 (R - a)$$

→ solving for $E_r(R)$: $2\pi L \varepsilon R E_r(R) = 2\pi L a(\rho_{sa} + \rho_0(R-a))$

$$E_r(r) = \frac{\rho_{sa}}{\varepsilon}\frac{a}{r} + \frac{\rho_0}{\varepsilon} a\left(\frac{R-a}{R}\right)$$

$$\vec{E}(\vec{r}) = \left(\frac{\rho_{sa}}{\varepsilon}\frac{a}{r} + \frac{\rho_0}{\varepsilon} a\left(\frac{R-a}{R}\right)\right)\hat{r}$$

b) Find the potential $V(\vec{r})$ in the region $a < r < b$.

Primary Equation $V(\vec{r}) = -\int \vec{E} \cdot d\vec{l}$

$$V(r) - \cancel{V(b)}^{0} = -\int_b^r \left(\frac{\rho_{sa}}{\varepsilon}\frac{a}{r} + \frac{\rho_0}{\varepsilon} a\left(1 - \frac{a}{r}\right)\right)\hat{r}\cdot\hat{r}\,dr$$

$$= -\frac{\rho_{sa}}{\varepsilon} a \ln(r/b) - \frac{\rho_0 a}{\varepsilon}(r-b) + \frac{\rho_0 a^2}{\varepsilon}\ln(r/b)$$

$$V(\vec{r}) = \frac{\rho_{sa} a}{\varepsilon}\ln(b/r) + \frac{\rho_0 a}{\varepsilon}(b-r) - \frac{\rho_0 a^2}{\varepsilon}\ln(b/r)$$

c) Find the potential difference between the conductors, $V_0 = V(a) - V(b)$. Then, express the surface charge density on the inner conductor, ρ_{sa}, as a function of V_0 and ρ_0.

$$V_0 = V(a) - V(b) = \frac{\rho_{sa} a}{\varepsilon}\ln(b/a) + \frac{\rho_0 a}{\varepsilon}(b-a) - \frac{\rho_0 a^2}{\varepsilon}\ln(b/a)$$

$$\rightarrow \rho_{sa} = \frac{\varepsilon}{a\ln(b/a)}\left(V_0 - \frac{\rho_0 a}{\varepsilon}(b-a) + \frac{\rho_0 a^2}{\varepsilon}\ln(b/a)\right)$$

$$\rho_{sa} = \frac{\varepsilon V_0}{a\ln(b/a)} - \frac{\rho_0}{\ln(b/a)}(b-a) + \rho_0 a$$

d) When a volumetric charge density exists in the region between two conductors, the capacitance cannot be computed using the usual formulas. Instead, it must be defined by $C = \frac{dQ}{dV_0}$, where Q is the total charge on one conductor and V_0 is the potential difference between the conductors. Using this definition and your results from earlier parts, determine the *capacitance per unit length* C' of the cable.

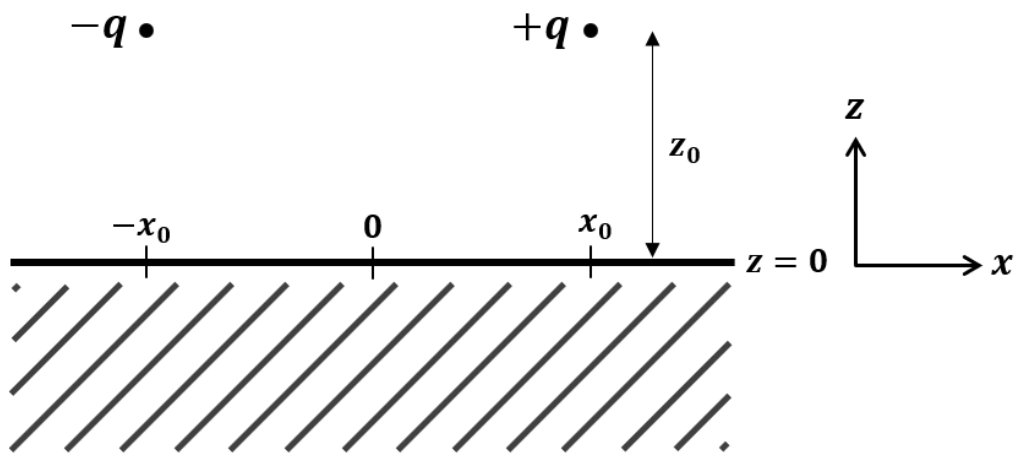
$$C' = \frac{1}{L}\frac{dQ}{dV_0} = \frac{1}{L}\frac{d}{dV_0}\left(2\pi L a\left(\frac{\varepsilon V_0}{a\ln(b/a)} - \frac{\rho_0}{\ln(b/a)}(b-a) + \rho_0 a\right)\right)$$

$$= \frac{1}{L}\cdot 2\pi L a \cdot \frac{\varepsilon}{a\ln(b/a)} = \frac{2\pi\varepsilon}{\ln(b/a)}$$

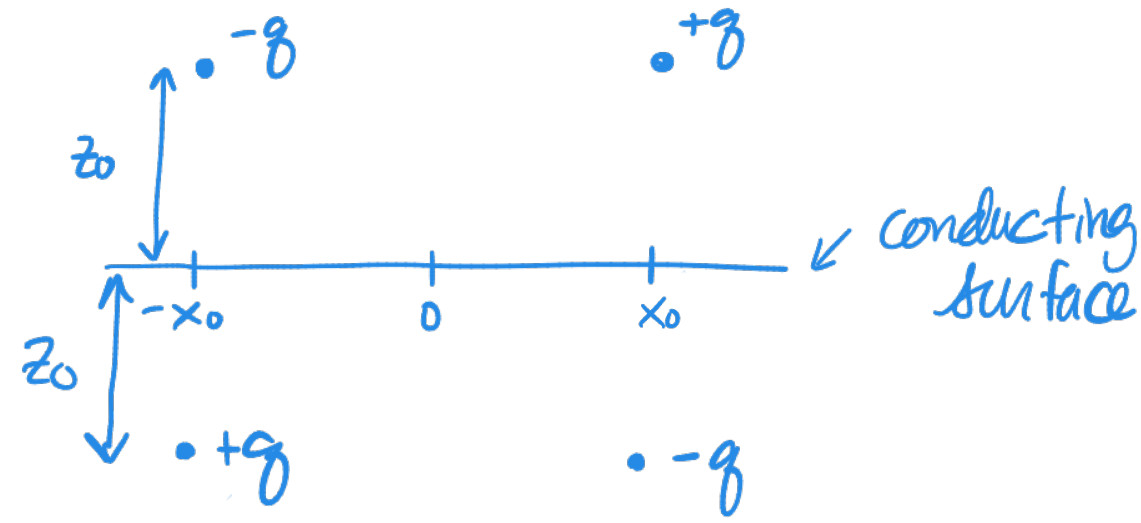
$$C' = \frac{2\pi\varepsilon}{\ln(b/a)}$$

Problem 4: Point Charges over a Conducting Plane [10 pts]

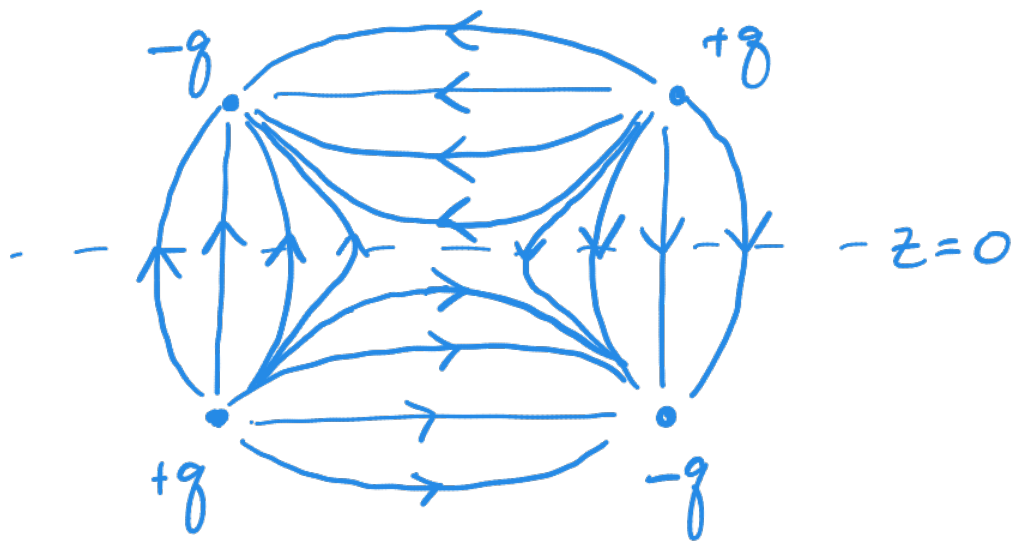
Two immobile point charges are held at a height z_0 above a perfectly conducting plane that occupies the region $z < 0$. A point charge $-q$ is located at $(-x_0, 0, z_0)$ and a charge $+q$ is located at $(x_0, 0, z_0)$. The region above the conductor ($z > 0$) has permittivity ε_0.



a) Draw a diagram in the x-z plane showing the geometry you will use for applying the method of images. Include the real charges, their image charges, their coordinates, and clearly mark the position of the conducting surface.

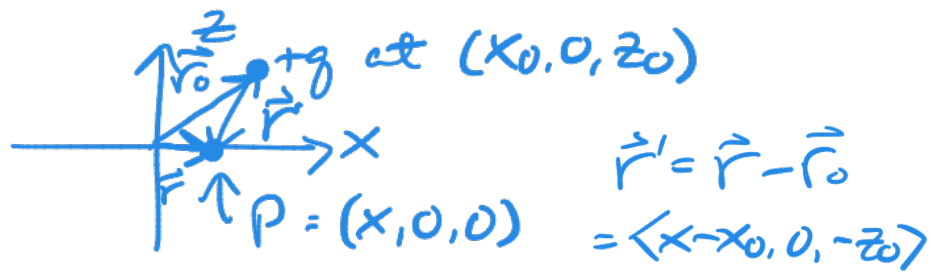


b) In a new diagram, sketch the electric field lines between the charges you drew in part a), including those in the region $z < 0$.



c) Determine the electric field $\vec{E}_{+q}$ at a point $P = (x, 0,0)$ solely due to the point charge $+q$ located at $(x_0, 0, z_0)$, ignoring the conducting plane.

Primary Equation $\vec{E}(\vec{r}) = \frac{q}{4\pi\varepsilon r^2}\hat{r}$



$\vec{r}' = \vec{r} - \vec{r}_0$

$= \langle x - x_0, 0, -z_0 \rangle$

$|\vec{r}'| = \sqrt{(x-x_0)^2 + z_0^2}$

$\hat{r}' = \frac{\vec{r}'}{|\vec{r}'|} = \frac{\langle x-x_0, 0, -z_0\rangle}{\sqrt{(x-x_0)^2 + z_0^2}} \rightarrow \vec{E}(\vec{r}) = \frac{q}{4\pi\varepsilon}\frac{\langle x-x_0, 0, -z_0\rangle}{((x-x_0)^2 + z_0^2)^{3/2}}$

$$\vec{E}_{+q}(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\frac{\langle x-x_0, 0, -z_0\rangle}{((x-x_0)^2 + z_0^2)^{3/2}}$$

d) Determine the total electric field $\vec{E}_{tot}$ at the point $P = (x, 0,0)$ on the surface of the conductor, taking into account both real charges and the effect of the conducting plane.

$$\left.\begin{aligned}
\vec{E}_{+q,real} &= \frac{q}{4\pi\varepsilon_0}\frac{\langle x-x_0, 0, -z_0\rangle}{((x-x_0)^2 + z_0^2)^{3/2}} \\
\vec{E}_{-q,real} &= -\frac{q}{4\pi\varepsilon_0}\frac{\langle x+x_0, 0, -z_0\rangle}{((x+x_0)^2 + z_0^2)^{3/2}} \\
\vec{E}_{+q,image} &= \frac{q}{4\pi\varepsilon_0}\frac{\langle x+x_0, 0, z_0\rangle}{((x+x_0)^2 + z_0^2)^{3/2}} \\
\vec{E}_{-q,image} &= -\frac{q}{4\pi\varepsilon_0}\frac{\langle x-x_0, 0, z_0\rangle}{((x-x_0)^2 + z_0^2)^{3/2}}
\end{aligned}\right\} E_{tot} = E_{+q,real} + E_{-q,real} + E_{+q,image} + E_{-q,image}$$

$$\vec{E}_{tot}(\vec{r}) = \frac{q}{4\pi\varepsilon_0}\left[\frac{\langle 0, 0, -2z_0\rangle}{((x-x_0)^2 + z_0^2)^{3/2}} + \frac{\langle 0, 0, 2z_0\rangle}{((x+x_0)^2 + z_0^2)^{3/2}}\right]$$

$$\vec{E}_{tot}(\vec{r}) = \frac{qz_0}{2\pi\varepsilon_0}\left[\frac{1}{((x+x_0)^2 + z_0^2)^{3/2}} - \frac{1}{((x-x_0)^2 + z_0^2)^{3/2}}\right]\hat{z}$$

e) Using your result from part d), calculate the surface charge density $\rho_s(\vec{r})$ on the conductor at the point $P = (x, 0,0)$.

Primary Equation	$D_{1n} - D_{2n} = \rho_s$

material 1 : air
material 2 : conductor $\rightarrow \varepsilon E_{1n} = \rho_s$

$$\rho_s = \varepsilon_0 \left[\frac{q}{4\pi\varepsilon_0} \left(2z_0 \left(\frac{1}{((x+x_0)^2 + z_0^2)^{3/2}} - \frac{1}{((x-x_0)^2 + z_0^2)^{3/2}} \right) \right) \right]$$

$$\rho_s(\vec{r}) = \frac{qz_0}{2\pi} \left[\frac{1}{((x+x_0)^2 + z_0^2)^{3/2}} - \frac{1}{((x-x_0)^2 + z_0^2)^{3/2}} \right]$$